Tristan Hoellinger | Double Master's degree in Mathematical Modelling and Hybrid AI

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Professional experience

Research Internship **Paris**
IAP - Institut d'Astrophysique de Paris *March 2023 - August 2023*
Intern in Cosmology in the "Large-scale structure and distant Universe" group at IAP. I work on novel simulation-based inference (SBI) techniques to optimally extract cosmological information from astronomical data.
Advisors : Florent Leclercq & Guilhem Lavaux

Master thesis **Châtillon**
ONERA - The French Aerospace Lab *February 2022 - Febryary 2023*
Intern in the NFLU (Digital methods for fluid mechanics) and MSAT (Advanced turbulence modeling and simulation) teams, in the DAAA departement. I develop new arbitrary high-order numerical methods to better preserve vortical structures in 3d turbulent flows simulation. Advisor : Jean-Baptiste Chapelier, jean-baptiste.chapelier@onera.fr

Assistant Bioinformatics Scientist **Toulouse**
Inserm, French National Institute for Health and Medical Research *September 2020 - January 2022*
I studied machine learning (XGBoost, deep neural networks) and heuristic methods to predict enhancer-gene relations in the human genome, based on high-throughput data. I presented my work at JOBIM conference 2021 (*Journées Ouvertes en Biologie, Informatique et Mathématiques* ; ranked 4th best poster among 46 for SFBI awards), and will soon publish my work as first author in a peer-reviewed journal article (currently being reviewed). Advisor : Sarah Djebali, sarah.djebali@inserm.fr

Research Internship **Barcelone (Espagne)**
CRG - Centre for Genomic Regulation *July 2021 - August 2021*
International summer internship in Guigó's lab at CRG Barcelona. Developpement of a Nextflow pipeline based on Surrogate Variable Analysis (SVA) for the dentification and analysis of genetic variants associated with intron retention estimated from RNA-sequencing in blood cell types from the Blueprint Project. Advisor : Diego Garrido : diego.garrido@crg.eu

Research Internship **Toulouse**
IMT (Toulouse Mathematics Institute) *July 2020*
Stochastic calculus. I studied recent theoretical developments about the convergence of repeated quantum non-demolition measurements. Advisors : Tristan Benoist, tristan.benoist@math.univ-toulouse.fr and Clément Pellegrini, clement.pellegrini@math.univ-toulouse.fr

Blibliographic Internship **Toulouse**
IRAP - Institut de recherche en astrophysique et planétologie *April 2020 - June 2020*
French laboratory of space astrophysics operated by CNRS (French National Centre for Scientific Research) and Paul Sabatier University. I studied kinetic scale plasma turbulence in the solar wind. Supervisor : Philippe Louarn, philippe.louarn@irap.omp.eu

Research Internship **Strasbourg**
Inria, National Institute for Research in Digital Science and Technology *juin 2019 - août 2019*

AI trainee in MIMESIS team. I analysed the robustness of a deep learning method for real-time data-driven biomechanical simulation, with respect to data sparsity and noises. As a result, we implemented an efficient transfer learning solution to learn new physical parameters with few data. The results were integrated in a peer-review journal article. Advisor : Andréa Mendizabal, andrea.mendizabal@inria.fr

Farm Hand **Saint-Broing-les-Moines**
La Ferme aux 100 Blés *July 2017*
Agricultural operation. I used various tools (pneumatic hammer, freight elevators, riveting guns), ran a market stall and delivered products.

Formation

INSA Toulouse, ENSEEIHT **Toulouse**
Double MS in Mathematics, Modelling and Artificial Intelligence *2020-2022*
I pursue a double-curriculum entitled "Mathematical Modelling and Artificial Intelligence", which leads to a Diplôme d'ingénieur (Master degree) in Mathematical Modelling awarded by INSA Toulouse (Institut National des Sciences Appliquées de Toulouse), and another Diplôme d'ingénieur in Computer Science and Telecommunications awarded by ENSEEIHT. Graduation in March 2023. Main (science) courses undertaken :

- smooth, non-smooth and stochastic optimization, (variational) data assimilation, optimal control
- signal and wavelets, advanced statistics, machine learning : clustering, kernel methods, CNN, recurrent networks, (variational) data assimilation networks, generative neural networks, reinforcement learning
- PDEs, finite volumes, discontinuous galerkine methods, model coupling, high performance computing, resolution of large linear systems, parallel programming

Paris-Saclay University **Gif-sur-Yvette**
Gap year and academic semester *2019-2020*
I took a year off to study physics (hamiltonian mechanics, quantum physics, mathematical physics, cosmology, electrodynamics, astrophysics, statistical physics) for one successful semester at the Magistère de physique d'Orsay, after what I completed my gap year with 2 internships in physics.

INSA Toulouse **Toulouse**
Bachelor of Science *2017–2019*
First 3 years of a 5-year course leading to a Diplôme d'ingénieur. Specialisation in Applied Mathematics.

INSA Strasbourg **Strasbourg**
Propaedeutic year, Ranked top 10% *2016–2017*
Basic sciences.

Lycée Charles-Emile Freppel **Obernai**
High school diploma, Science, Highest Honors (18,1/20) *2013–2016*

Student projects

- Variational data assimilation to estimate river bathymetry from altimetric data.
- Kalman filters and ensemble Kalman filters : theory and application to linear and non-linear cases.
- Machine learning-based method to predict the energy of a molecule based on its geometric structure. Several approaches implemented, such as randomized Coulomb matrices and *Wavelet Scattering Transform*.
- Implementation of a Variational Assimilation Network, with PyTorch, for supervised learning of Observed Dynamical Systems. Application to a simple linear dynamical system and to a Lorenz 40d system.

Languages

French: Mother Tongue **German**: basics (A2)

English: Fluent. Cambridge Linguaskill Business : CERCL C1+ (highest score, 180+/180)

Concrete Skills in Applied Mathematics

Modelling and numerical simulation: CFD, solving PDEs, ODE, discontinuous galerkine methods, finite volume methods, lagrangian methods

Statistics: inferential statistics, model selection, parametric tests, non-parametric tests, meta-modelling, gaussian processes, bayesian methods

Generalities: Optimal control, signal and image processing, smooth non-convex and convex non-smooth optimization, stochastic optimization

Machine learning: clustering, kernel methods, bagging, boosting, neural networks (perceptrons, CNNs, recurrent networks, adversarial networks), variational data-assimilation networks, reinforcement learning

High Performance Computing: solving large sparse systems, parallel computing (OpenMP, MPI)

Concrete Skills in Computer Science

Programmation: C++, Fortran, Python (pandas, PyTorch, Tensorflow, JAX), Julia, R, bash & Awk, OCaml

Version: Git, Github

Others: OpenMP, MPI, Docker, Kubernetes, AWS

Simulation: Python, Maple, Matlab

Office: Suite Office, LaTeX, LibreOffice

Web: HTML5, CSS, Jekyll

Miscellaneous

- Former Mentor of 8 High School students for 6 month, in the context of the Equal Opportunity Program « Ô Talents », former President of the Theatre Troupe "La Catin"
- Scholarship holder during my studies (level 6/7)
- Hobbies : photography, sport (martial arts, running, hiking, bouldering)

Publications

- As co-author : Data-driven simulation for augmented surgery, Andrea Mendizabal et al., 2020, Developments and Novel Approaches in Biomechanics and Metamaterials. Advanced Structured Materials, vol 132. Springer, Cham. `https://doi.org/10.1007/978-3-030-50464-9_5`
- As principal author : The genome-wide identification of enhancer/gene relationships : new reliable benchmark datasets needed to make progress in the field, T. Hoellinger, S. Djebali (currently being reviewed, available on bioRxiv `https://doi.org/10.1101/2022.10.12.511908`)

References

Internship Advisors

Sarah Djebali, permanent researcher at INSERM,
sarah.djebali@inserm.fr

Jean-Baptiste Chapelier, permanent at ONERA,
jean-baptiste.chapelier@onera.fr

Professors

Violaine Roussier-Michon, lecturer (Maîtresse de Conférence) at INSA Toulouse,
roussier@insa-toulouse.fr